INCH-POUND

MIL-C-43191G
INTERIM AMENDMENT 2 (GL)
15 October 1990
SUPERSEDING
INTERIM AMENDMENT 1 (GL)
7 September 1990

MILITARY SPECIFICATION

CLOTH, WIND RESISTANT SATEEN, COTTON AND NYLON

This interim amendment is approved for use within the Department of the Army, with MIL-C-43191G dated 16 October 1989.

PAGE 1

1.2, under "Class 3": Add the following:

* "Class 4 - Desert camouflage printed (6 color) and Quarpel treated
* Class 5 - Desert camouflage printed (3 color) and Quarpel treated"

PAGE 2

2.1.2, under "2-1-1516B - Woodland Pattern - 60 inches": Add the following:

* "2-1-1884 - 6 Color Desert Pattern 48 & 60 inch pattern (class 4)
* 2-1-2240 - 3 Color Desert Pattern 48 & 60 inch pattern (class 5)"

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After TABLE II: Add the following new paragraphs:

"3.4.2.3 Class 4. The cloth shall be dyed to a ground shade approximating Light Tan 379. The Desert Camouflage pattern shall be obtained by roller or screen printing using the five remaining rollers or screens, as appropriate, for Tan 380, Light Brown 381, Dark Brown 382, Black 383 and Khaki 384 (see 6.5.3).

* 3.4.2.3.1 Class 5. The cloth shall be dyed to a ground shade approximating Light Tan 492. The three color Desert Camouflage pattern shall be obtained by roller or automatic screen printing using two or three rollers or screens as appropriate for the Light Tan 492, Light Brown 493 and Light Khaki 494 areas of the pattern. The use of two rollers or screens is appropriate when the ground shade matches Light Tan 492. The dyeing of the ground shade (Light Tan 492) and the printing of the cloth shall be accomplished with organic colorants (see 6.5.4) to provide a match to each of the three shades of the pattern and to provide the spectral reflectance levels specified in Table IIB. Each dyed and printed color area shall match the specific color on the standard sample (see 6.3). Resin bonded pigment printing will not be permitted.

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* 3.4.2.4 Pattern execution classes 4 and 5. The pattern shall reproduce the standard sample in respect to design, colors and registration of the respective areas. The warpwise pattern repeat of the dyed, printed and finished cloth shall be 16.75 +1.25 -1.75 inches. Each pattern area shall show solid coverage. Skitteriness exceeding that shown by the standard sample in any of the printed areas will not be acceptable. When the standard sample is not referenced for pattern execution, a pattern drawing shall be provided and the pattern on the finished cloth shall match that of drawing 2-1-1884 for class 4 and 2-1-2240 for class 5 (see 2.1.2, 6.2 and 6.4).

3.4.2.5 Spectral reflectance, class 4. The spectral reflectance factors for the 6 color Desert Camouflage printed finished cloth shall conform to the requirements specified in Table IIA when tested as specified in 4.4.3.

TABLE IIA. Spectral reflectance requirements, class 4

Wavelengths nanometers	Reflectance values (percent)					
	Light Tan 379		Tan 380 and Khaki 384		Light Brown 381 and Dark Brown 382	
	(min)	(max)	(min)	(max)	(min)	(max)
700	38	53	25	44	19	41
720	38	54	25	45	20	41
740	39	55	25	46	20	42
760	40	56	26	47	21	42
780	41	57	27	48	21	42
800	43	58	28	50	22	43
820	45	59	30	52	23	45
840	48	62	33	55	24	46
860	50	65	36	58	25	48

* 3.4.2.6 Spectral reflectance, class 5. The spectral reflectance factors for the 3 color Desert Camouflage printed finished cloth shall conform to the requirements specified in Table IIB when tested as specified in 4.4.3.

TABLE IIB. Spectral reflectance requirements, class 5

Wavelengths nanometers	Reflectance values (percent)					
	Light Tan 492		Light Brown 493		Light Khaki 494	
	(min)	(max)	(min)	(max)	(min)	(max)
700	38	53	19	41	25	44
720	38	54	20	41	25	45
740	39	55	20	42	25	46
760	40	56	21	42	26	47
780	41	57	21	42	27	48
800	43	58	22	43	28	50
820	45	59	23	45	30	52
840	48	62	24	46	33	55
860	50	65	25	48	36	58"

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Add the following new paragraph:

* "3.4.5.3 Classes 4 and 5. The desert camouflage printed and water repellent treated cloth shall show fastness to light, laundering (after 3 cycles) and perspiration equal to or better than the standard sample or equal to or better than a rating of "good" for each of the pattern areas when tested as specified in 4.4.3. The class 4 cloth shall show fastness to crocking equal to or better than the standard sample or shall have and AATCC Chromatic Transference Scale rating of not less than 3.5 for all the pattern areas, except Black 383 which shall have an AATCC Chromatic Transference Scale rating of not lower than 1.5. The class 5 cloth shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating of not less than 3.5 for all the pattern areas. Testing shall be as specified in 4.4.3."

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* 3.6, line 3: Delete "class 2 and 3" and substitute "classes 2, 3, 4 and 5".

* 3.10, paragraph identifier: Delete "(classes 2 and 3)" and substitute "(classes 2, 3, 4 and 5)".

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* 3.11, paragraph identifier: Delete "(classes 2 and 3)" and substitute "(classes 2, 3, 4 and 5)".

* 3.12, paragraph identifier: Delete "(classes 2 and 3)" and substitute "(classes 2, 3, 4 and 5)".

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* 4.4.2.1.1, lines 11, 12, 16, 18, 20, 22 and 26: Delete "(class 3)" and substitute "(classes 3, 4 and 5)".

* Lines 23, 24 and 25: Delete "(classes 2 and 3)" and substitute "(classes 2, 3, 4 and 5)".

* After line 14: Add new defect as follows:

"Pattern repeat less than 15.00 inches or more than 18.00 inches (classes 4 and 5)".

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TABLE V: Delete characteristic for "Spectral reflectance" and substitute the following in their respective columns:

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PAGES 12 and 13

"Spectral reflectance			
	Class 2	3.4.1.1	4.5.1
	Class 3	3.4.2.2	4.5.1
	Class 4	3.4.2.5	4.5.1
*	Class 5	3.4.2.6	4.5.1"

* After characteristics for: "Hydrostatic pressure, Dynamic absorption, Air permeability, Water repellents, Resistance to organic liquids and Water repellency": Delete "(classes 2 and 3)" and substitute "(classes 2, 3, 4 and 5)".

PAGE 14

* 4.5.1, line 2, after "(nm)": Insert "for class 3 and 700 nm to 860 nm for classes 4 and 5".

* Line 14, after "the standard.": Insert "For classes 4 and 5 Desert camouflage, specimens shall be measured as a single layer backed with two layers of the same shade cut from the standard. Light Tan 379 shall be measured using small area of view."

* Delete the last sentence and substitute the following:

"When the measured reflectance values for any color at four or more of the listed wavelengths do not meet the limits for the Woodland pattern (class 3) in table II, 6 color Desert pattern (class 4) in table IIA, and 3 color Desert pattern (class 5) in table IIB, it shall be considered a test failure."

PAGE 15

6.2, "f": Delete and substitute:

"f. Woodland or applicable Desert camouflage pattern drawing if required (see 3.4.2.1 or 3.4.2.4)."

PAGE 16

6.4: Delete the first sentence and substitute the following:

"For access to the standard shade sample (see 3.2) and the Woodland camouflage pattern drawing or appropriate Desert camouflage pattern drawing, if applicable (see 3.4.2.1 or 3.4.2.4) address the contracting activity issuing the invitation for bids or request for proposal."

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After 6.5.2: Add the following new paragraph:

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"6.5.3 Dye formulation, class 4. The areas of the pattern have been found to be satisfactory when dyed and printed with various combinations of the following dyes for 6 colored areas:

Light Tan 379 (Ground Shade)
Vat Brown 1
Vat Black 25

Tan 380
Vat Yellow 1
Vat Yellow 33
Vat Brown 1
Vat Brown 57
Vat Black 25

Light Brown 381
Vat Yellow 1
Vat Yellow 33
Vat Brown 1
Vat Brown 57

Dark Brown 382
Vat Yellow 1
Vat Yellow 33
Vat Brown 1
Vat Brown 57

Black 383
Sulfur Black 6/Vat Black 14
Cibanone Black R

Khaki 384
Vat Yellow 1
Vat Yellow 33
Vat Brown 1
Vat Black 25

Satisfactory dyeing of the Light Tan 379, ground shade, and printing of Tan 380, Light Brown 381, Dark Brown 382 and Khaki 384 has been accomplished by the use of vat dyes only. Black 383 has been satisfactorily printed with a combination of vat and sulfur dyes."

* 6.5.4 Dye information, class 5. The Light Tan 492 (Ground Shade), Light Brown 493 and Light Khaki 494 areas of the 3 color Camouflage pattern have been found to be satisfactory when dyed or printed with the various combinations of dyes for Light Tan 379, Light Brown 381 and Khaki 384 respectively as listed in 6.5.3."

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Preparing activity:

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